

SEMESTER 7

Advanced Cryptography

Course Code	HNCS709	CIE Marks	40
Teaching Hours/Week (L:T:P)	3:0:0	ESE Marks	60
Credits	3	Exam Hours	2 Hrs. 30 Min.
Prerequisites (if any)		Course Type	Theory

Course Objectives:

1. To equip the learner in the theoretical foundations of quantum mechanics and quantum algorithms, such as Shor's and Grover's, in order to evaluate their destabilizing impact on classical public-key cryptography and understand the principles of quantum key distribution.
2. To investigate the mathematical hard problems and design principles underlying post-quantum cryptographic families—specifically hash-based, code-based, lattice-based, and multivariate systems—to develop secure alternatives resilient against quantum adversaries.

SYLLABUS

Module No.	Syllabus Description	Contact Hours
1	Quantum Computing-Introduction, Quantum mechanics Shor's Factoring Algorithm-Factoring, Reduction from factoring to period-finding, Shor's period-finding algorithm, Continued fractions Grover's Search Algorithm, Quantum Cryptography-Saving cryptography from Shor, Quantum key distribution, Reduced density matrices and the Schmidt decomposition, The impossibility of perfect bit commitment, More quantum cryptography (Quantum Computing: Lecture Notes- Ronald de Wolf)	9
2	Introduction to post-quantum cryptography- Is cryptography dead?, A taste of post-quantum cryptography, Challenges in post-quantum cryptography, Comparison to quantum cryptography Classical Cryptography and Quantum computing, The computational model, The quantum Fourier transform, The hidden subgroup problem	9
3	Hash-based Digital Signature Schemes-Hash based one-time signature schemes, Merkle's tree authentication scheme, One-time key-pair generation using an PRNG, Authentication path computation, Tree chaining, Distributed signature generation, Security of the Merkle Signature Scheme Code-based cryptography-Introduction, Cryptosystems, The security	9

	of computing syndromes as one-way function, Codes and structures	
4	Lattice-based Cryptography-Introduction, Preliminaries, Finding Short Vectors in Random q -ary Lattices, Hash Functions, Public Key Encryption Schemes, Digital Signature Schemes Multivariate Public Key Cryptography-Introduction, The Basics of Multivariate PKCs, Examples of Multivariate PKCs, Basic Constructions and Variations, Standard Attacks (module 2-4 -Post-Quantum Cryptography by Daniel J. Bernstein, Johannes Buchmann, Erik Dahmen)	9

Course Assessment Method
(CIE: 40 marks, ESE: 60 marks)

Continuous Internal Evaluation Marks (CIE):

Attendance	Assignment/ Microproject	Internal Examination-1 (Written)	Internal Examination- 2 (Written)	Total
5	15	10	10	40

End Semester Examination Marks (ESE)

In Part A, all questions need to be answered and in Part B, each student can choose any one full question out of two questions

Part A	Part B	Total
2 Questions from each module. - Total of 8 Questions, each carrying 3 marks (8x3 =24marks)	- Each question carries 9 marks. - Two questions will be given from each module, out of which 1 question should be answered. - Each question can have a maximum of 3 sub divisions. (4x9 = 36 marks)	60

Course Outcomes (COs)

At the end of the course students should be able to:

Course Outcome		Bloom's Knowledge Level (KL)
CO1	Develop problem-solving and analytical skills in quantum computation	K3
CO2	Describe the foundations and goals of post-quantum cryptography	K2
CO3	Apply authentication mechanisms in hash-based systems and understand the foundations of code-based cryptography	K3
CO4	Apply lattice-based cryptographic constructions and Multivariate Public Key Cryptography for data security	K3

Note: K1- Remember, K2- Understand, K3- Apply, K4- Analyse, K5- Evaluate, K6- Create

CO-PO Mapping Table:

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11
CO1	3	2	2								2
CO2	3										2
CO3	3	2	2								2
CO4	3	2	2								2

Text Books				
Sl. No	Title of the Book	Name of the Author/s	Name of the Publisher	Edition and Year
1	Post-Quantum Cryptography	Daniel J. Bernstein · Johannes Buchmann Erik Dahmen	Springer	1/e, 2009
2	Quantum Computing	Ronald de Wolf	QuSoft, CWI and University of Amsterdam	I/e, 2023

Reference Books				
Sl. No	Title of the Book	Name of the Author/s	Name of the Publisher	Edition and Year
1				
2				
3				

Video Links (NPTEL, SWAYAM...)	
Module No.	Link ID
1	
2	
3	
4	

MODEL QUESTION PAPER
APJ ABDUL KALAM TECHNOLOGICAL UNIVERSITY SEVENTH SEMESTER B. TECH HONOURS DEGREE EXAMINATION, MONTH AND YEAR
Course Code: